SONiX 32-Bit Cortex-M0 Micro-Controller

CMP Example Code

AMENDENT HISTORY

Version	Date	Description
1.0	2024/11/26	1. First version.

Table of Content

1 OVERVIEW

This document provides our customers the C program examples of RAIL TO RAIL COMPARATOR.

2 FEATURE

- Rail-to-Rail structure comparators.
- Use ARM-compiler V6.12

3 HW

- The starter kit board of vendor desired MCU.
- Connect CMnP0 / CMnN0 / OP0O with desire voltage source.

4 EXAMPLE CODE

4.1 Initialization

Initial CMP function. User can modify the settings of CMP in this function. Depending on the chip series, the source selection for the positive and negative input functions may be different.

```
//------------------------
//User Code starts HERE!!!
CMP_init();
```

```
/*****************************************************************************
* Function    : CMP_init
* Description : Initialization of CMP0/CMP1/CMP2/CMP3
* Input       : None
* Output      : None
* Return      : None
* Note        : None
*****************************************************************************/
void CMP_init(void)
{
  __CMP_ENABLE_CMPHCLK;

#if CMP_VIREF_ENABLE
  __ADC_ENABLE_HCLK
  CMP_VIREF_Enable(0x7F);
#endif

#if CMP0_ENABLE
  SN_CMP->CTRL |= (mskCMP_CM0PS_VIREF0 | mskCMP_CM0O_ENABLE | mskCMP_CM0G_FALLING);
  SN_CMP->IE |= mskCMP_CM0IE_ENABLE;
  NVIC_EnableIRQ(CMP0_IRQn);
  __CMP0_ENABLE;
#endif

#if CMP1_ENABLE
  SN_CMP->CTRL |= (mskCMP_CM1PS_VIREF0 | mskCMP_CM1O_ENABLE | mskCMP_CM1G_FALLING);
  SN_CMP->IE |= mskCMP_CM1IE_ENABLE;
  NVIC_EnableIRQ(CMP1_IRQn);
  __CMP1_ENABLE;
#endif

#if CMP2_ENABLE
  SN_CMP->CTRL1 |= (mskCMP_CM2PS_VIREF0 | mskCMP_CM2O_ENABLE | mskCMP_CM2G_FALLING);
  SN_CMP->IE |= mskCMP_CM2IE_ENABLE;
  NVIC_EnableIRQ(CMP2_IRQn);
  __CMP2_ENABLE;
#endif

#if CMP3_ENABLE
  SN_CMP->CTRL1 |= (mskCMP_CM3PS_VIREF0 | mskCMP_CM3O_ENABLE | mskCMP_CM3G_FALLING);
  SN_CMP->IE |= mskCMP_CM3IE_ENABLE;
  NVIC_EnableIRQ(CMP3_IRQn);
  __CMP3_ENABLE;
#endif
}
```

5 RESULT

The results of CMP can be observed through CMnP0 / CMnN0 / OP0O pins. Comparator compares positive terminal’s voltage and negative terminal’s voltage, and then output result to output pin. When V+ >V-, comparator outputs high status. When V+ < V-, comparator outputs low status.

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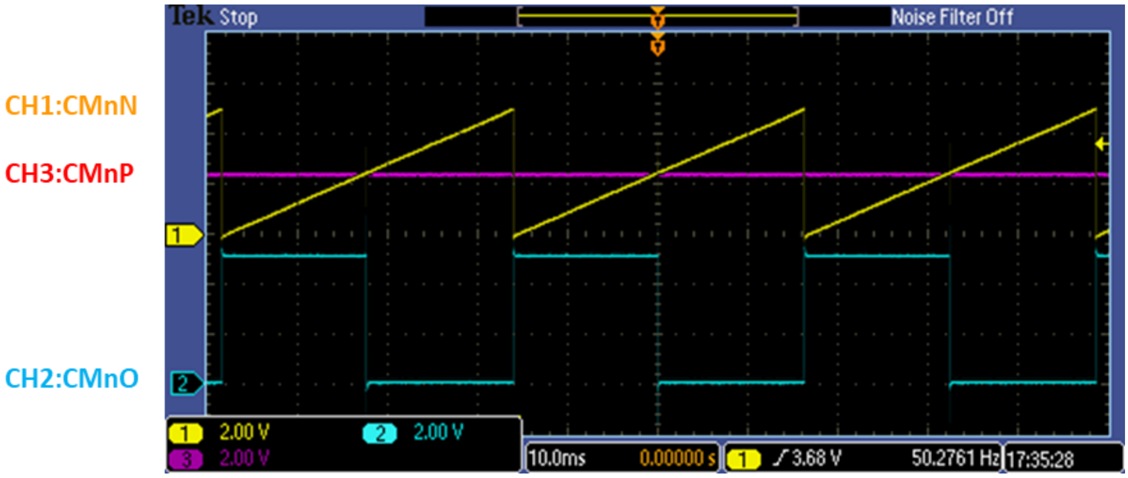

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